Awan, S. E., Bennamoun, M., Sohel, F., Sanfilippo, F., & Dwivedi, G. (2022). "A reinforcement learning-based approach for imputing missing data." *Neural Computing and Applications, 34*(12), 9701–9716. https://doi.org/10.1007/s00521-022-06958-3. The article is a peer-reviewed journal article and was published by Springer in 2022.

Domain/Application Area: Data preprocessing and missing-data management in machine learning and data analytics.

Specific ML Problem: Estimating missing numerical values in tabular datasets while preserving the natural variability of the original data.

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Task 1: Article Summary

1(a) Citation, Domain and ML Problem

The selected article was written by **Saqib Ejaz Awan, Mohammed Bennamoun, Ferdous Sohel, Frank Sanfilippo, and Girish Dwivedi** and published in *Neural Computing and Applications* in 2022. Springer identifies it as an original article, with publication on 11 February 2022.

The research belongs to the domain of **machine learning data preprocessing**, specifically **missing-value imputation**. Missing values can reduce the quality of statistical analysis and machine-learning models. Conventional univariate methods such as mean or median imputation replace all missing values in a feature with a single value, which can reduce the variance of the completed dataset. The authors therefore investigate whether **reinforcement learning (RL)** can learn a better strategy for estimating missing numerical values while retaining variability.

1(b) Objective and Research Gap — Summary

The main objective of the article is to develop a reinforcement-learning method for imputing missing numerical values without unnecessarily reducing the variability of the original data. The authors identify a gap in conventional univariate single-imputation methods: mean, median and similar techniques use essentially one representative value for missing entries in a column, which can distort the feature distribution and underestimate variance. Existing multivariate machine-learning methods such as k-nearest neighbours and Random Forests can use relationships between features, but they may require greater computational effort and can introduce their own limitations. The authors propose a different approach in which a Q-learning agent learns a policy for increasing or decreasing an initial estimate until it reaches a lower-error state. Unlike conventional univariate methods, the proposed method can produce different estimates for different missing entries in the same column. The researchers evaluate the approach on eight UCI datasets and compare it with statistical, machine-learning, and generative imputation techniques. Their goal is therefore not simply to reduce numerical error, but also to maintain the natural distribution and variability of the completed data.

Task 2: Methodology Analysis

2(a) ML Technique and Dataset

Reinforcement Learning Technique

The authors use **Q-learning**, a model-free reinforcement-learning technique. The problem is formulated as a **Markov Decision Process (MDP)** with states, actions, rewards and transitions. The agent maintains a Q-table and updates it according to the standard Q-learning update rule:

$$\mathbf{Q(s,a) \leftarrow Q(s,a) + \alpha[R(s,a) + \gamma \max Q(s',a) - Q(s,a)]}$$

The authors define **10 states**, where a state represents how far the current estimate is from the desired low-error region. There are only **two actions**: increase the current estimated value or decrease it. An epsilon-greedy strategy is used during learning so that the agent initially explores different actions and gradually shifts toward exploitation.

The imputation process initially generates an estimate, such as a column mean. The RL policy then repeatedly modifies this estimate. The objective is to move toward the state associated with minimum imputation error. The authors report using 10,000 Q-learning iterations in the experimental setup and repeating the imputation 100 times to investigate generalizability.

Dataset

The researchers use **eight publicly available datasets from the UCI Machine Learning Repository**.

Dataset	Instances	Attributes	Application/Feature Type
Breast Cancer	569	30	Digitized cell-nucleus characteristics such as radius, texture and perimeter
Vehicle	946	18	Vehicle silhouette features
Travel	980	10	TripAdvisor customer-feedback features
Spambase	4,601	57	Email and word/character-frequency features
Parkinson Disease	5,875	19	Voice-measurement features
Letter Recognition	20,000	17	Features describing images of capital letters

Default Credit Card	30,000	24	Credit, age and payment-history variables
News Popularity	39,644	61	Online-news statistics

The dataset information is reported by the authors and reproduced in the article's Table 7.

For the experiments, the authors artificially introduce missing values at **5%, 10%, 15%, and 20%**. The missing entries are randomly generated and therefore represent a **Missing Completely At Random (MCAR)** setting. The datasets are divided into **70% training and 30% testing** portions. The missing values are represented as NaN, and Q-learning hyperparameters are selected using grid search. Performance is measured using **Mean Absolute Error (MAE)** and **Root Mean Squared Error (RMSE)**.

Benchmark Methods

The RL approach is compared against:

- Mean imputation
- Median imputation
- Most-frequent-value imputation
- k-Nearest Neighbours
- Random Forest
- Multiple Imputation by Chained Equations (MICE)
- GAIN
- CGAIN

This provides a useful comparison between simple statistical methods, traditional ML methods and more advanced neural/generative methods.

2(b) Mapping to CO 4

The article maps most directly to **Reinforcement Learning**, one of the CO 4 topics.

CO 4 Topic: Reinforcement Learning

The mapping is clear because the algorithm contains the standard RL components:

State → Action → Reward → New State → Policy

The Q-table represents the learned action-value function, while epsilon-greedy exploration allows the agent to discover an effective imputation strategy. The article therefore gives a practical example of how a classical RL algorithm can be applied outside traditional game-playing or robotics applications.

Methodological Strength

One major strength is that the method can generate **different imputed values for different missing entries within the same feature**. This addresses a major weakness of mean/median imputation, which tends to replace every missing value with the same estimate. The authors also test the method across eight datasets and multiple missing-data proportions, making the evaluation broader than a single-dataset experiment.

Methodological Limitation

A significant limitation is that the experiments assume **MCAR missingness**. Real-world missing data may instead be Missing At Random (MAR) or Missing Not At Random (MNAR). The paper itself also identifies a limitation that the current approach works with **numeric variables only**, and the authors propose extending it to categorical variables in future work.

Another important methodological concern is that the RL state is defined using the error relative to the ground-truth value. This is practical for controlled experiments where the original value is known, but genuine missing values in production do not have an accessible ground truth at prediction time. Consequently, further validation is needed to demonstrate how the policy can be applied when the missing value itself is genuinely unknown.

Task 3: Results & Critical Evaluation

3(a) Key Results

The authors report that the proposed RL approach outperformed the competing approaches on **six of the eight datasets** and remained in the top three on the other two datasets. Performance was evaluated primarily using lower MAE and RMSE values.

A representative comparison for the **Spambase dataset** is given below.

Comparison at 5% Missing Data

Method	MAE	RMSE
Mean	0.0271	0.0544
Median	0.0201	0.0591
Most Frequent	0.0208	0.0615
kNN	0.0309	0.0719
Random Forest	0.0309	0.0696

MICE	0.0286	0.0588
GAIN	0.0501	0.0723
CGAIN	0.0447	0.0611
Proposed RL	**0.0183**	**0.0485**

The proposed approach achieves the lowest MAE and RMSE in this comparison.

Comparison at 20% Missing Data

Method	MAE	RMSE
Mean	0.0278	0.0593
Median	0.0210	0.0635
Most Frequent	0.0217	0.0667
kNN	0.0319	0.0750
Random Forest	0.0321	0.0715
MICE	0.0290	0.0620
GAIN	0.0595	0.0764
CGAIN	0.0430	0.0601
Proposed RL	**0.0198**	**0.0527**

Even with 20% missing data, the proposed method remains the best-performing method on Spambase according to both reported metrics.

The authors also repeated the Spambase experiment 100 times. For 5%, 10%, 15%, and 20% missingness, the reported mean MAE values were **0.01781, 0.01859, 0.01980, and 0.02017**, respectively. The corresponding RMSE values were **0.04936, 0.05012, 0.05100, and 0.05287**.

The researchers also found that the imputed data could preserve the original distribution relatively well, particularly at lower missing-data rates.

3(b) Critical Evaluation

The reported results are promising, but they should not automatically be interpreted as evidence that Q-learning is universally superior to other imputation methods.

First, the use of **MAE and RMSE is appropriate** for numerical imputation because both directly measure the difference between an imputed value and its known ground truth. RMSE additionally penalizes large errors more strongly.

However, the evaluation has an important limitation: missing values are created **randomly**, meaning the experiment represents MCAR conditions. Real-world missingness can be systematic. For example, a user may avoid providing income information precisely because the income is high, or a medical measurement may be missing because a patient has a particular condition. Such MAR or MNAR mechanisms can be substantially more difficult.

There is also an evaluation-protocol concern. Because the researchers possess the original values, they can calculate the error of an estimate against the ground truth while developing the RL environment. In real deployment, the true missing value is unavailable. Although the learned policy can theoretically be transferred to unseen cases, this distinction should be investigated more explicitly.

Another concern is that the paper does not demonstrate the method across a broad set of **real missingness mechanisms** or categorical variables. The authors themselves identify numeric-only data as a limitation.

Experiments That Would Strengthen the Findings

A stronger study would include:

1. **MAR and MNAR experiments** in addition to MCAR.
2. Comparisons with newer imputation methods such as modern transformer-based or diffusion-based models.
3. Statistical significance testing across multiple random seeds rather than relying primarily on average error.
4. An ablation study examining the effect of the number of RL states, learning rate, discount factor and exploration schedule.
5. Evaluation of downstream impact—for example, whether imputing with the RL method improves classification accuracy when the completed data are subsequently used by a classifier.
6. Experiments using categorical and mixed numerical-categorical datasets.

3(c) Real-World Applicability

The method could be useful in applications where incomplete numerical data frequently occur. Examples include healthcare records, financial datasets, customer analytics, industrial monitoring, sensor systems and online-business data.

For example, in healthcare, patient measurements may be missing because certain tests were not performed. In finance, information about customers may be incomplete. In industrial environments, sensors may temporarily fail. An imputation approach that preserves variability instead of replacing all missing values with one average could produce more realistic datasets for subsequent analytics.

However, industry deployment presents several challenges. The first is **data availability**: a reliable RL policy may require historical datasets where the true values are known during training. The second is **distribution shift**: a policy learned from one population may not work effectively on another population. The third is **different missingness mechanisms**. Industry data may contain MAR or MNAR patterns rather than purely random missingness. Another challenge is computational and operational complexity compared with simple mean or median imputation.

There are also **ethical concerns**, particularly in healthcare and financial applications. Incorrectly imputed values can influence medical decisions, credit assessments or customer profiling. Therefore, imputation uncertainty and provenance should be tracked rather than treating every generated value as ground truth.

Task 4: Reflection & Learning Outcome

4(a) Three Key Insights

Insight 1: Reinforcement Learning Is a General Decision-Making Framework

The article demonstrates that RL is not limited to games or robotics. The same fundamental structure of **state, action, reward and policy** can be used to solve a data-preprocessing problem.

This deepens the CO 4 concept of **Reinforcement Learning** because it shows how an RL agent can learn a sequence of actions that gradually improves an objective.

Insight 2: The Reward Function Determines What the Agent Learns

The paper shows that defining the reward structure is central to RL. The agent learns to move toward a minimum-error state because the reward matrix encourages the appropriate transitions. Therefore, an RL algorithm is not useful simply because Q-learning is used; the environment and reward formulation must correctly represent the actual problem.

This connects directly with the CO 4 concept of **MDPs and reward-based learning**.

Insight 3: Model Performance Depends on the Evaluation Protocol

The experiments demonstrate that the RL method performs strongly under the MCAR setting, but this does not guarantee equal performance under MAR or MNAR conditions.

This connects to the broader CO 4 learning objective of understanding that **machine-learning performance depends on data assumptions, training conditions and evaluation methodology**, not merely on the choice of algorithm.

4(b) Proposed Extension / Novel Experiment

A useful extension would be to develop a **hybrid Q-learning imputation system for mixed numerical and categorical datasets under MAR and MNAR conditions**.

The new experiment would extend the state and action spaces so that the agent could handle both numerical adjustment and categorical selection. Instead of assuming that missing values occur randomly, missingness could be generated according to realistic relationships with other variables. The experiment could then compare the original Q-learning approach with the enhanced agent, Random Forest, MICE, GAIN and a modern neural imputation model.

This extension is feasible because the original paper already provides the basic Q-learning framework, and its authors explicitly identify categorical variables and additional environmental information as future research directions.

The expected impact would be significant because such a system would move the method closer to real-world datasets, where missingness is rarely perfectly random and mixed data types are common. It would also provide a stronger test of whether RL genuinely provides an advantage over simpler statistical imputation methods.